# Algebraic Geometry WS 26/27

RECAP Sheet 5: 9.11-13.11

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This sheet is for the week of 9.11 to 13.11, where there will be no lectures or presentation sessions. It is meant to help you catch up on Category Theory, which will be used throughout the lectures.

The book Category theory in Context, by Emily Riehl, is an excellent resource for this and this sheet accompanies Chapter 1 (except Section 1.7), as well as Sections 3.1-3.4 of Chapter 3. These sections contain the prerequisite material needed for the lectures.

## R — Recall

1. What is a category ? A functor ? A natural transformation ?
2. What is a criterion for a natural transformation to be a natural isomorphism ?
3. What is a criterion for a functor to be an equivalence of categories ?
4. What is a limit cone ? Why do limits exist in the category of sets, and how are they described ?
5. What is a pullback ? An equalizer ?
6. Describe pullbacks in $\mathrm{Set}, \mathrm{Ab}, \mathrm{CRing}$ and revisit the proof that the structure sheaf of a commutative ring is indeed a sheaf.

## E — Explore

1. Give examples of categories, functors and natural transformations. Try to give your own examples.
2. For every example of a category that you know, does it have limits ? And for every functor that you know, does it preserve limits ?
3. How does a poset give rise to a category ?
4. What are some limits of rings, besides the ones appearing in the structure sheaf of a given ring ?
5. Describe pushouts in Set, as well as colimits indexed by the poset $\mathbb{N}$. Find your own examples of indexing categories and describe colimits indexed by those.

## C — Connect

1. How does the definition of a presheaf relate to the notion of a functor ? How does one concretely describe a presheaf ?
2. Let $\mathcal{F}$ be a presheaf. How do the conditions "$\mathcal{F}$" is a sheaf and "$\mathcal{F}$ preserves limits" relate ?

## C — Connect (continued)

3. How do the notions of limits and colimits relate ?
4. What is a contravariant functor, and how does it relate to a functor ?

## A — Apply

1. Let $G$ be a group, and $BG$ denote the category with one object $*$ and $\hom_{BG}(*,*) = G$, with composition given by the multiplication of $G$.
   (a) Prove that functors $BG \to \mathrm{Set}$ correspond to actions of $G$ on sets, and that natural transformations correspond to maps that are compatible with this action.
   (b) Deduce a description of limits and colimits indexed by $BG$.
2. Let $F, G : C \to D$ be functors and $\eta : F \to G$ a natural transformation. Let $X : I \to C$ be a diagram. Assume $X$ admits a limit preserved by both $F$ and $G$. Prove that if each $\eta_{X(i)} : F(X(i)) \to G(X(i))$ is an isomorphism, then so is $\eta_{\lim_I X} : F(\lim_I X) \to G(\lim_I X)$.
3. (a) What are monomorphisms in the category of abelian groups ? Of commutative rings ?
   (b) Prove that in a category $C$, a morphism $f : x \to y$ is a monomorphism if an only if the map from $x$ to the diagram $x \to y \leftarrow x$ is a limit diagram, i.e. if and only if $x$ is a pullback $x \times_y x$.
   (c) Deduce that limit-preserving functors preserve monomorphisms.
   (d) Deduce a description of monomorphisms of sheaves.
4. Prove that a pullback $A \to B \leftarrow C$ of sets, where both maps are inclusions of subsets, is simply the intersection $A \cap C$.
5. Consider the limit of abelian groups $\lim_n \mathbb{Z}/p^n\mathbb{Z}$ taken along the projections $\mathbb{Z}/p^{n+1}\mathbb{Z} \to \mathbb{Z}/p^n\mathbb{Z}$. Find a concrete description and prove that it is torsion-free.
6. Exercise 1.2.iv in Category theory in Context
7. Exercises 3.1.iv to 3.1.vii in Category theory in Context.
8. Prove that pushouts in CRing are given by relative tensor products.